# HUST

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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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# MAXIMUM A POSTERIOR


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# Introduction

## Statistical Estimation & MAP Framework

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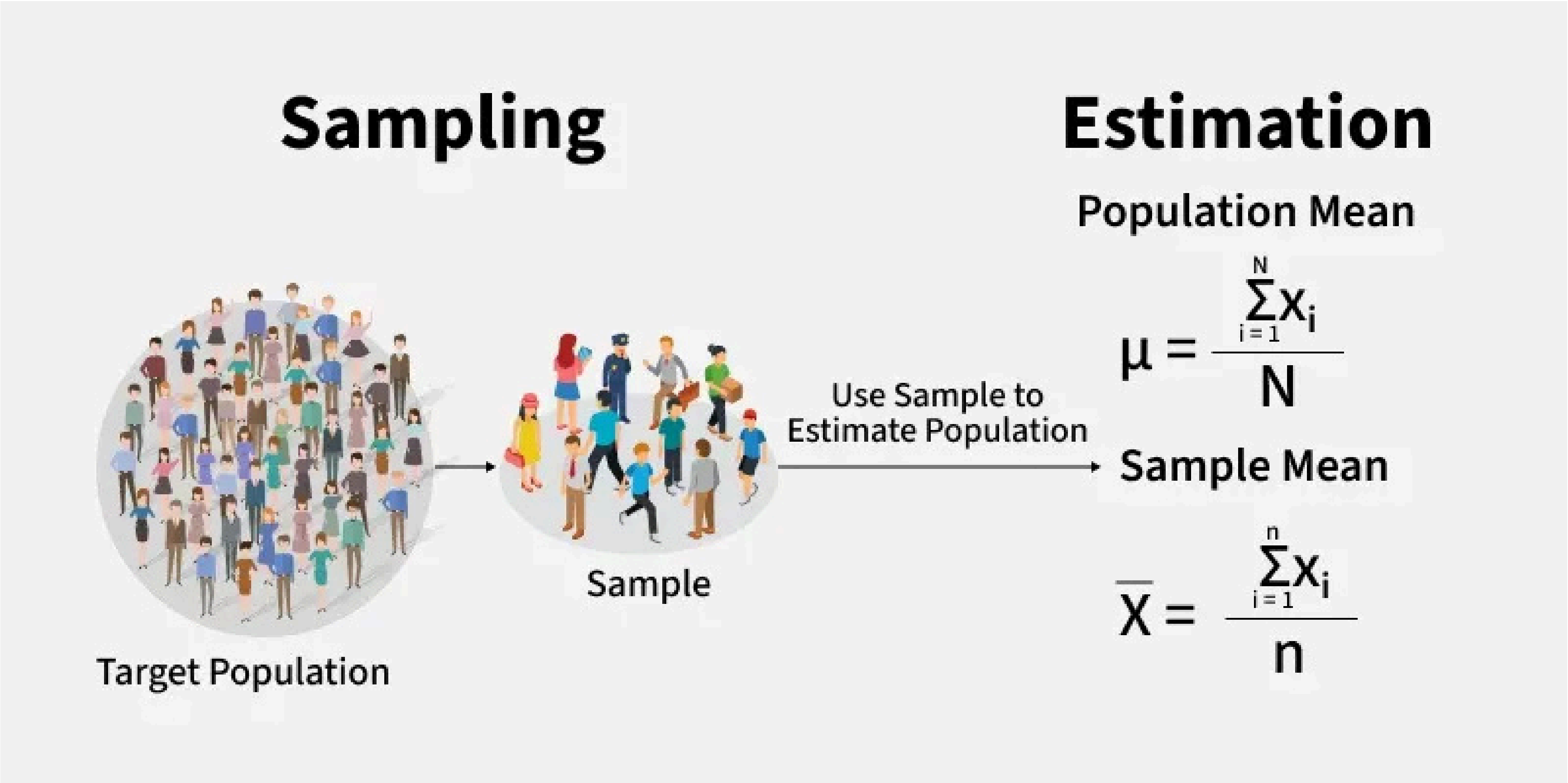
# What is Statistical Estimation?




- **Goal:**

Given observed data, find the parameter θ that best explains it.

- **Core question:**

*"Which value of θ fits the data most?"*

- **Two main schools of thought:**
  - **Frequentist →** Maximum Likelihood Estimation (MLE)
  - **Bayesian →** Maximum A Posteriori (MAP), Full Bayesian

# MAP Estimation

- **Bayes' Theorem:**

$$\underbrace{p(\theta|\boldsymbol{y})}_{\text{Posterior}} = \frac{\underbrace{p(\theta)}_{\text{Prior}}\underbrace{p(\boldsymbol{y}|\theta)}_{\text{Likelihood}}}{\underbrace{p(\boldsymbol{y})}_{\text{Marginal likelihood}}}$$

- **MAP objective:** Find θ that maximizes the posterio

$$\theta_{MAP} = \arg\max_{\theta} p(\theta|y) = \arg\max_{\theta} \log p(\theta|y) = \arg\max_{\theta} \log \frac{p(\theta)p(\boldsymbol{y}|\theta)}{p(\boldsymbol{y})}$$

- **Intuition:** Balance between *evidence* from *data* and *prior belief* about θ

- **Why log?** Convert products → sums
  - Avoid numerical underflow
  - Easier to differentiate

# Project Concept

**GOAL**

Estimate the Maximum A Posteriori (MAP) value of insurance claim using machine learning approaches that combine prior knowledge and observed data.

## PROBLEM OVERVIEW

Given historical insurance data and prior information, the project estimates the most probable claim amount (MAP) using different modeling approaches.

## KEY APPROACHES

**Coin Flipping (Baseline)**
Simple probabilistic baseline using prior likelihood.

**Sensor Fusion**
Combine multiple data sources to improve estimation.

**Linear Regression**
Predict claim value using linear regression model.

## WORKFLOW

**Input Data**
Historical claims + prior knowledge (sensor data, etc.)

→

**Models**
- Coin Flipping
- Sensor Fusion
- Linear Regression

→

**MAP Estimation**
Compute MAP estimate for claim amount

→

**Evaluation**
Evaluate and compare model performance

## EXPECTED OUTCOMES

- Improve estimation accuracy
- Compare performance of different approaches
- Provide insights for insurance decision-making

# Implementation Details

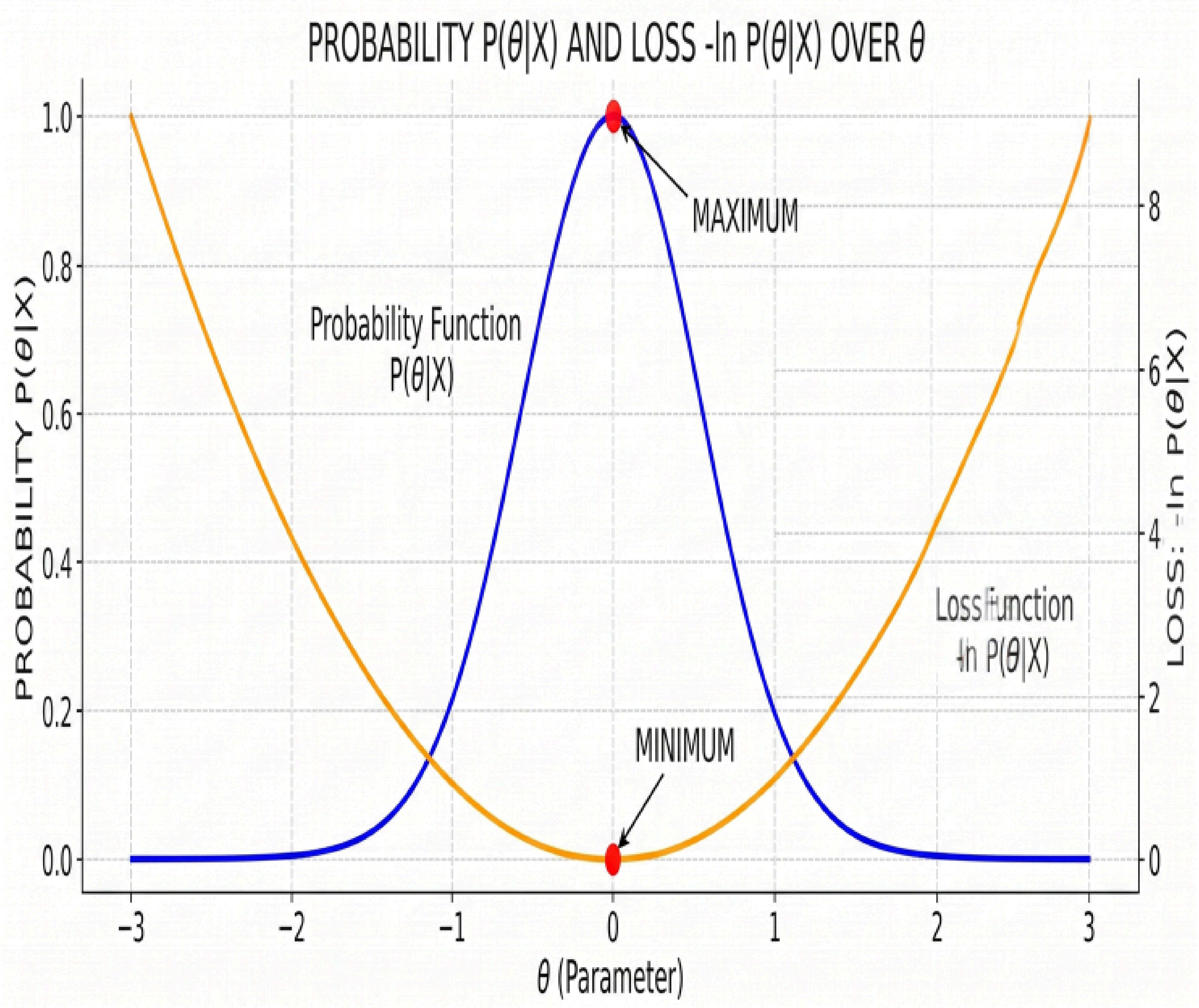
# Mathematical Transformation - Why use Gradient Descent?

- The objective of MAP (Maximum A Posteriori): Find the parameter θ that maximizes the posterior probability $P(\theta|X)$

- Problem transformation: From Maximize Probability →Minimize Negative Log-Posterior (Loss Function).

- Decomposition formula:

$$\text{Loss}(\theta) = -\log P(X|\theta) - \log P(\theta)$$



# Building the Optimizer (GradientDescentOptimizer)

- **Hyperparameters:** learning_rate, ($\alpha$)epochs, tol (convergence threshold).
- **Update Rule:**

$$\theta_{new} = \theta_{old} - \alpha \cdot \nabla_{\theta \text{Loss}}(\theta)$$

- **Convergence Check:** $||\theta_{new} - \theta_{old}|| < \text{tol}$

```python
import numpy as np

class GradientDescentOptimizer:
    def __init__(self, learning_rate=0.01, epochs=1000, tol=1e-6):
        self.lr = learning_rate
        self.epochs = epochs
        self.tol = tol

    def minimize(self, theta_init, grad_func, data, **kwargs):
        """
        General Gradient Descent Algorithm.
        Goal: Find theta to minimize the loss function (Negative Log-Posterior).
        """
        theta = np.array(theta_init, dtype=float)

        for i in range(self.epochs):
            # Compute the gradient at the current point
            grad = grad_func(theta, data, **kwargs)

            # Update parameters (stepping in the opposite direction of the gradient)
            new_theta = theta - self.lr * grad

            # Check for convergence
            if np.linalg.norm(new_theta - theta) < self.tol:
                break
            theta = new_theta

        return theta
```

# Calculating Gradients for 3 Examples (Gradients Implementation)

- **Example 1 (Coin Flipping):** Bernoulli + Beta. Technical note: np.clip the parameter θ within the range $(0,1)$ to avoid division-by-zero errors.

- **Example 2 (Sensor Fusion):** Gaussian Gaussian. The derivative is the linear sum of the Likelihood residual and the Prior residual.

- **Example 3 (Ridge Regression):** Linear + Gaussian.

$$\nabla_w \text{Loss} = \frac{X^T(Xw - y)}{\sigma^2} + \lambda w$$

```python
# Clip theta within (0, 1) to avoid log errors or division by zero
theta = np.clip(theta, 1e-10, 1 - 1e-10)
```

```python
    # Derivative of the -Log Likelihood (Gaussian)
    grad_likeli = np.sum(theta - data) / (sigma_likeli**2)
    # Derivative of the -Log Prior (Gaussian)
    grad_prior = (theta - mu_prior) / (sigma_prior**2)

    return grad_likeli + grad_prior
```

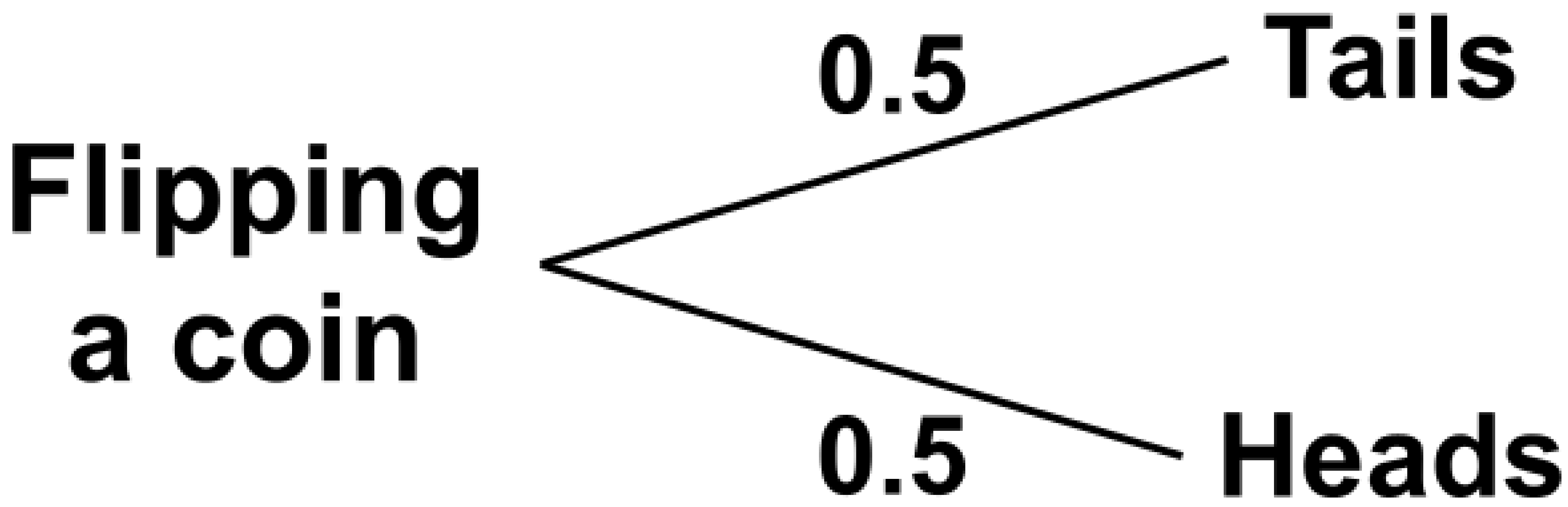
```python
# Add a column of ones to account for the bias term (intercept w0)
X_b = np.column_stack([np.ones(len(X)), X])

# Ridge derivative formula: X.T @ (Xw - y) / sigma^2 + lambda * w
prediction_error = X_b @ w - y
gradient = (X_b.T @ prediction_error) / (sigma**2) + lmbda * w
return gradient
```

# Example 1

## Compare MAP and MLE through the coin flipping problem

# Coin Flipping Problem




- We flip a coin **n** times
- Each result is represented as: 1 = Head; 0 = Tail
- The goal is to estimate: **θ** = probability of getting Head

# Method Ideas

1. **Maximum Likelihood Estimation (MLE)**

- MLE estimates the parameter using only the observed data.
- For the Bernoulli coin flipping problem:

$$\theta_{MLE} = \frac{k}{n}$$

**with k = Number of heads**
**n = Observations**

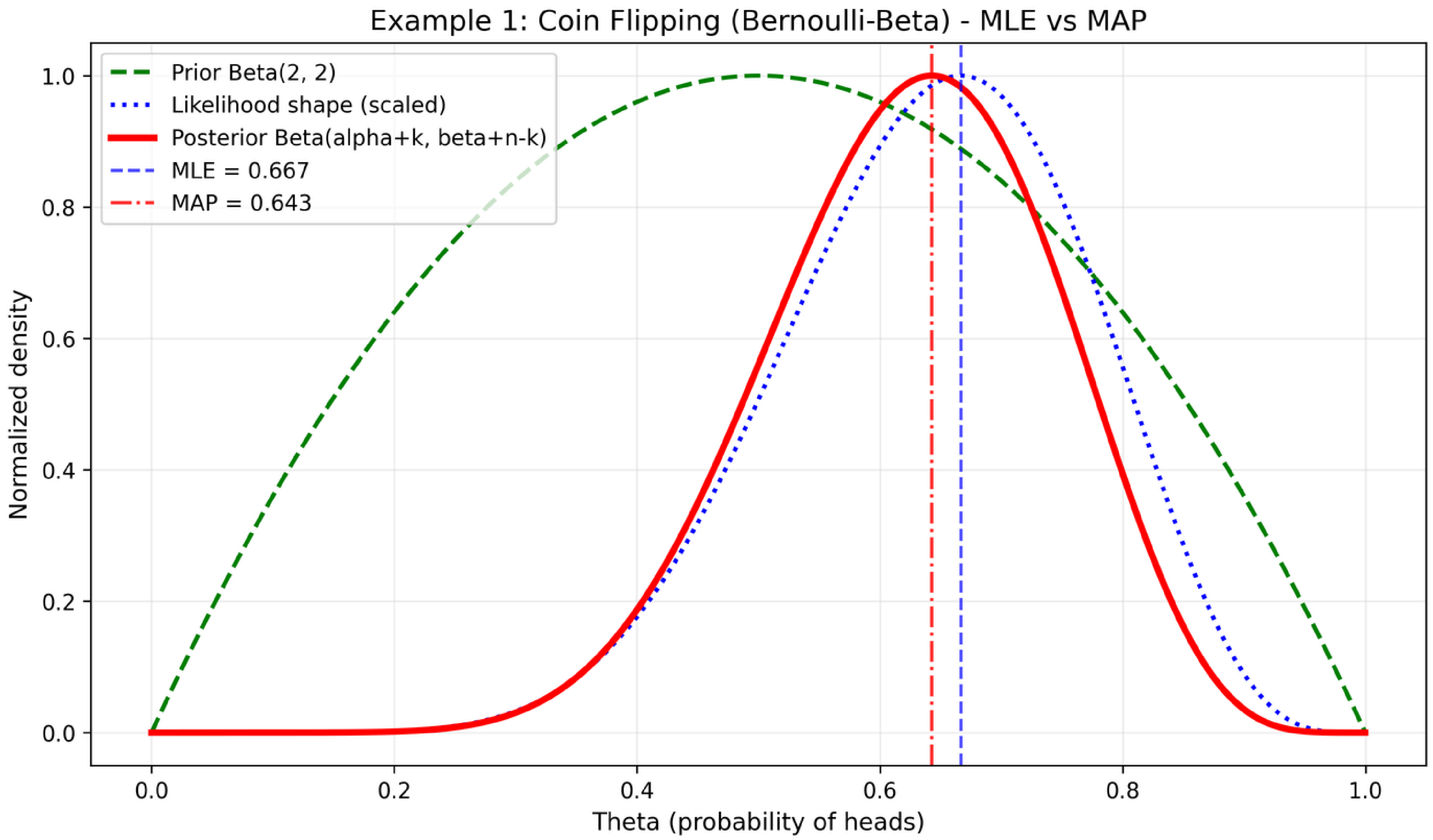
2. **Maximum A Posteriori Estimation (MAP)**

- MAP combines: **Observed data + Prior belief**
- We use a Beta prior: $\theta \sim Beta(\alpha, \beta)$
- The MAP formula:

$$\theta_{MAP} = \frac{k + \alpha - 1}{n + \alpha + \beta - 2}$$

# Visual Comparison

- Observations (n): 12
- Heads (k): 8
- Tails (n-k): 4
- Beta prior: α = 2, β = 2




- The MLE estimate is closer to the observed data.
- The MAP estimate is influenced by both the data and the prior belief.
- Because the prior Beta(2,2) favors values near 0.5, MAP is slightly pulled toward the center.

# Pros/ Cons between MLE and MAP

| | MLE | MAP |
|---|---|---|
| **Pros** | • Very simple; often closed-form and fast.<br>• Uses only observed data, easy to interpret. | • Combines data evidence with prior belief.<br>• More robust in low-data settings (regularization effect). |
| **Pros** | • Can be unstable with small samples.<br>• No mechanism to include prior domain knowledge. | • Requires selecting a prior (can introduce bias if poorly chosen).<br>• May require numerical optimization in complex models. |

## Conclusion:

- **MLE is suitable when we have a large amount of data.**
- **MAP is useful when the dataset is small or when prior knowledge is available.**

# Example 2

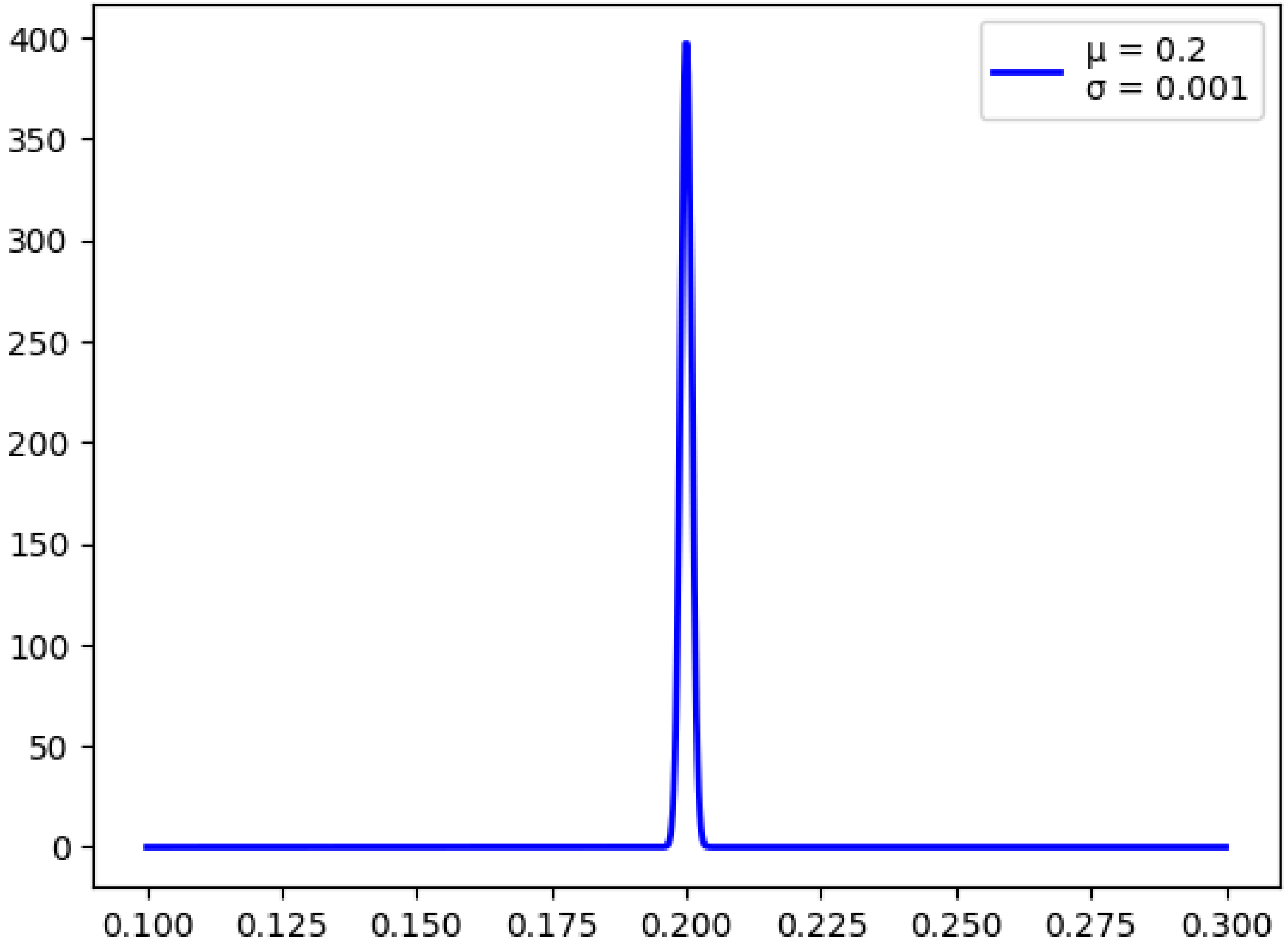
## High-Precision Sensor Calibration in Noisy Environments

# Problem Statement - Systemic Bias Estimation

| Temperature | Sensor1 | Sensor2 | Sensor3 | Sensor4 | Sensor5 |
|---|---|---|---|---|---|
| 25,43149 | 0,677916 | 0,208399 | 0,033098 | 0,145758 | 0,291933 |
| 24,48606 | 0,801851 | 0,445989 | 0,869082 | 0,499015 | 0,53149 |
| 20,05509 | 0,781817 | 0,040596 | 0,205773 | 0,517225 | 0,598457 |
| 22,78169 | 0,65103 | 0,316018 | 0,945797 | 0,433242 | 0,378673 |
| 25,29283 | 0,292357 | 0,893463 | 0,289188 | 0,490347 | 0,012502 |

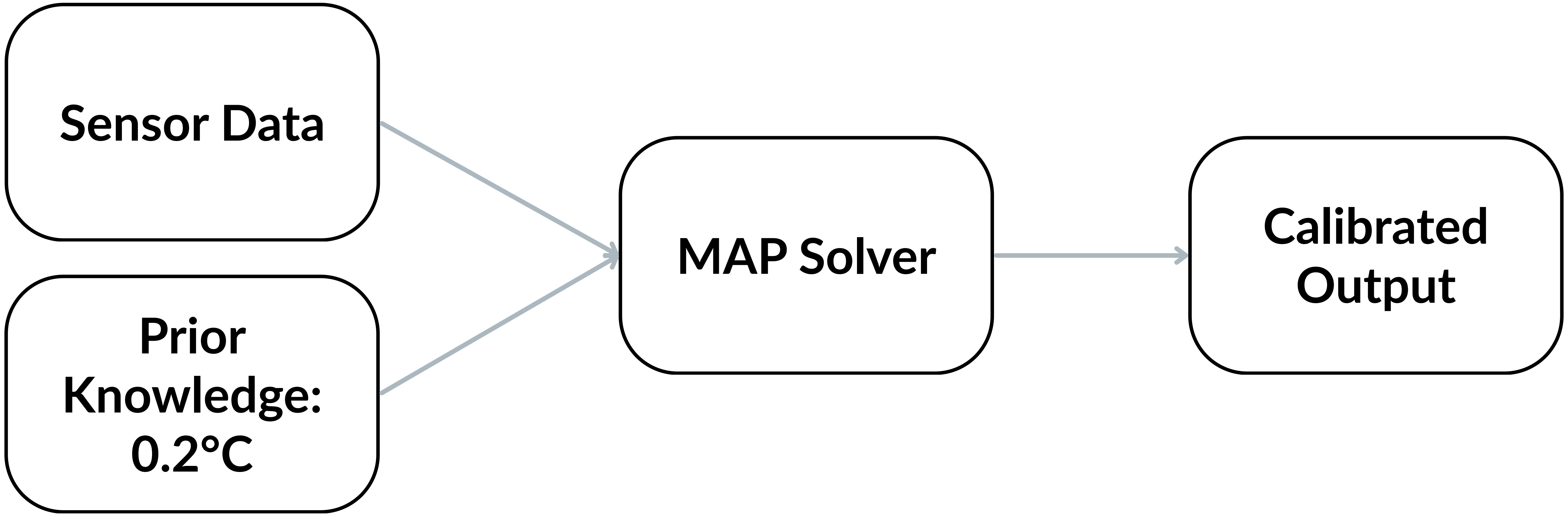
- Objective: Identify fixed systemic bias for precise sensor calibration.
- Dataset: 17,000 records from 5 independent sensors.
- Challenge: Stochastic noise makes simple MLE (averaging) unreliable.

# Problem Statement - Systemic Bias Estimation



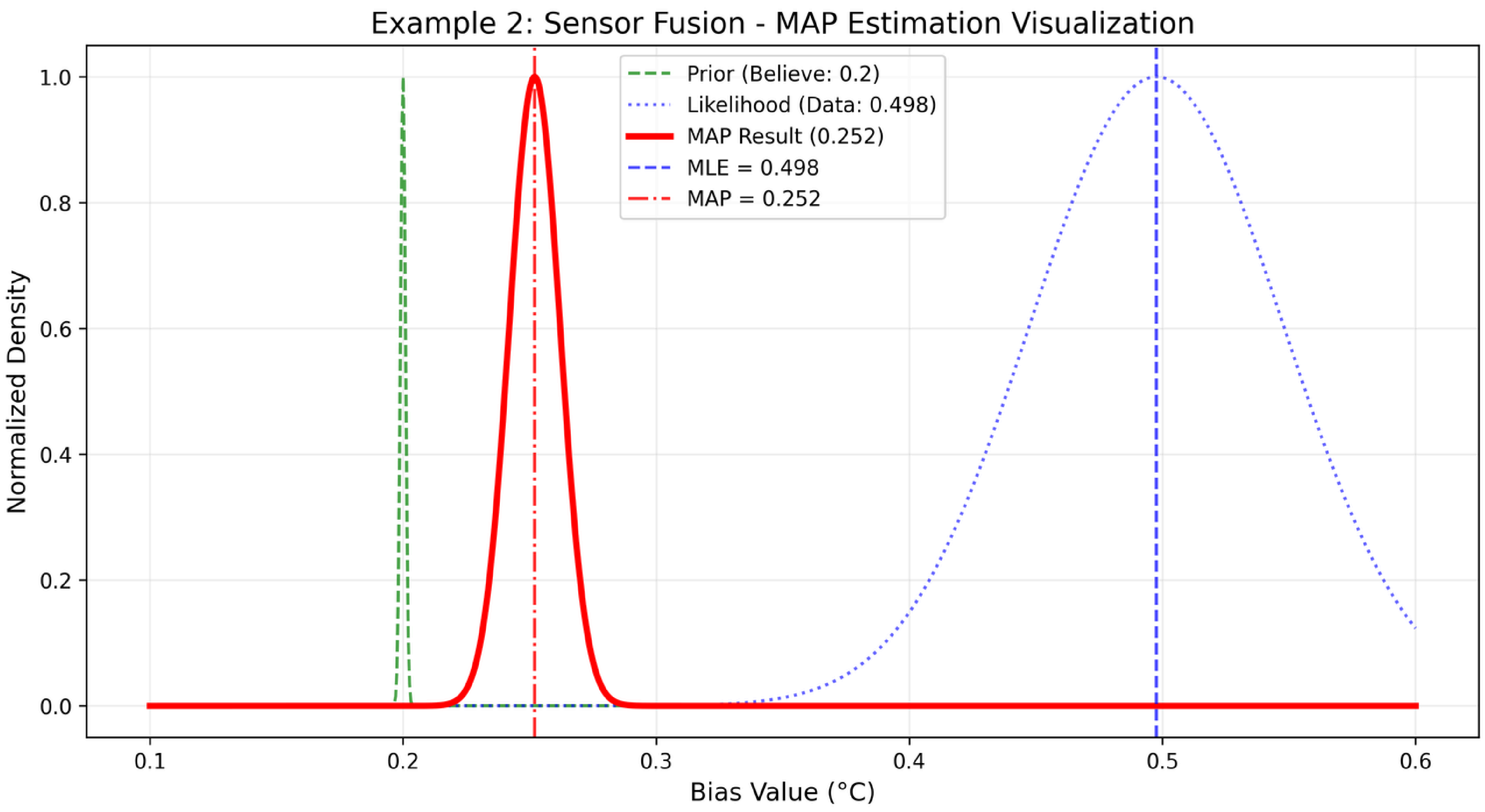

- Prior Assumption ($\mu_{prior}$): Fixed at +0.2°C, representing high-end sensor technology.
- Confidence Level ($\sigma_{prior}$): Set at 0.001, reflecting absolute trust in technical specs.
- Significance: Acts as a mathematical anchor to resist transient measurement noise.

# MAP Framework & Implementation




- Probabilistic Model: Log-Gaussian Likelihood for sensor noise modeling.
- Optimization Strategy: Custom Gradient Descent for objective minimization.
- System Integration: Modular pipeline for automated data processing.

# Visual Analysis: MLE vs. MAP Results




- Likelihood (Blue): Peaks at 0.498°C — Severe MLE bias due to noisy data.
- MAP (Red): Peaks at 0.252°C — Optimal balance via expert prior.
- Core Takeaway: MAP filters environmental noise to stabilize the system near the 0.2°C baseline.

# Example 3

## Predicting parameters in linear regression

# Problem Statement - Weights Estimation

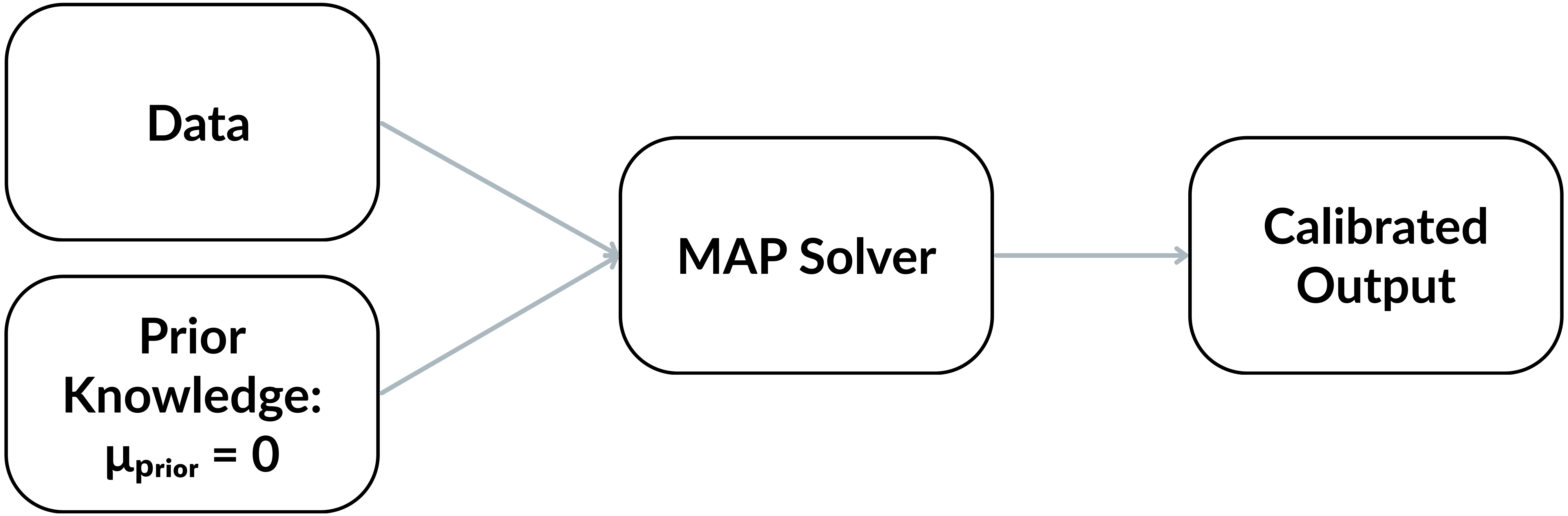
| 1 | age | sex | bmi | children | smoker | charges | region_northwest | region_southeast | region_southwest | charges_log |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | -1.440417729 | 0 | -0.453159594 | -0.90923416 | 1 | 16884.924 | FALSE | FALSE | TRUE | 9.734176431773115 |
| 3 | -1.51164747 | 1 | 0.50942164544 | -0.07944162 | 0 | 1725.5523 | FALSE | TRUE | FALSE | 7.453302452133774 |
| 4 | -0.799350061 | 1 | 0.38315460035 | 1.5801434662 | 0 | 4449.462 | FALSE | TRUE | FALSE | 8.400538468975023 |
| 5 | -0.443201357 | 1 | -1.305052191 | -0.90923416 | 0 | 21984.471 | TRUE | FALSE | FALSE | 9.998091601725266 |
| 6 | -0.514431098 | 1 | -0.292456082 | -0.90923416 | 0 | 3866.8552 | TRUE | FALSE | FALSE | 8.260196845858372 |
| 7 | -0.585660839 | 0 | -0.807363253 | -0.90923416 | 0 | 3756.6216 | FALSE | TRUE | FALSE | 8.231275322 |
| 8 | 0.48278527438 | 0 | 0.45530719754 | -0.07944162 | 0 | 8240.5896 | FALSE | TRUE | FALSE | 9.016827173741753 |
| 9 | -0.158282394 | 0 | -0.479396902 | 1.5801434662 | 0 | 7281.5056 | TRUE | FALSE | FALSE | 8.893092932994646 |
| 10 | -0.158282394 | 1 | -0.136672066 | 0.7503509232 | 0 | 6406.4107 | FALSE | FALSE | FALSE | 8.765054439884247 |
| 11 | 1.48000164671 | 0 | -0.790964936 | -0.90923416 | 0 | 28923.137 | TRUE | FALSE | FALSE | 10.272397139270046 |
| 12 | -1.013039284 | 1 | -0.728651329 | -0.90923416 | 0 | 2721.3208 | FALSE | FALSE | FALSE | 7.908872629665523 |
| 13 | 1.62246112847 | 0 | -0.717172507 | -0.90923416 | 1 | 27808.725 | FALSE | TRUE | FALSE | 10.233105102955316 |
| 14 | -1.155498766 | 1 | 0.61273104596 | -0.90923416 | 0 | 1826.843 | FALSE | FALSE | TRUE | 7.510344619461671 |
| 15 | 1.19508268319 | 0 | 1.50151985686 | -0.90923416 | 0 | 11090.718 | FALSE | TRUE | FALSE | 9.313863803 |

- Objective: Predicting parameters for linear regression analysis of the problem.
- Dataset: Information of 1337 households (standardized).
- Challenge: Stochastic noise makes simple MLE (averaging) unreliable.

# Problem Statement - Weights Estimation

- Prior Assumption ($\mu_{prior}$): Set at 0, assuming the features have very little impact on the result.
- Confidence Level ($\sigma_{prior}$): Set at 0.4, reflecting belief in a priori knowledge.
- Significance: acts as a neutral prior belief that features should not strongly influence the target unless supported by data, while simultaneously serving as a mathematical anchor to resist transient measurement noise by constraining coefficients toward zero.

# MAP Framework & Implementation

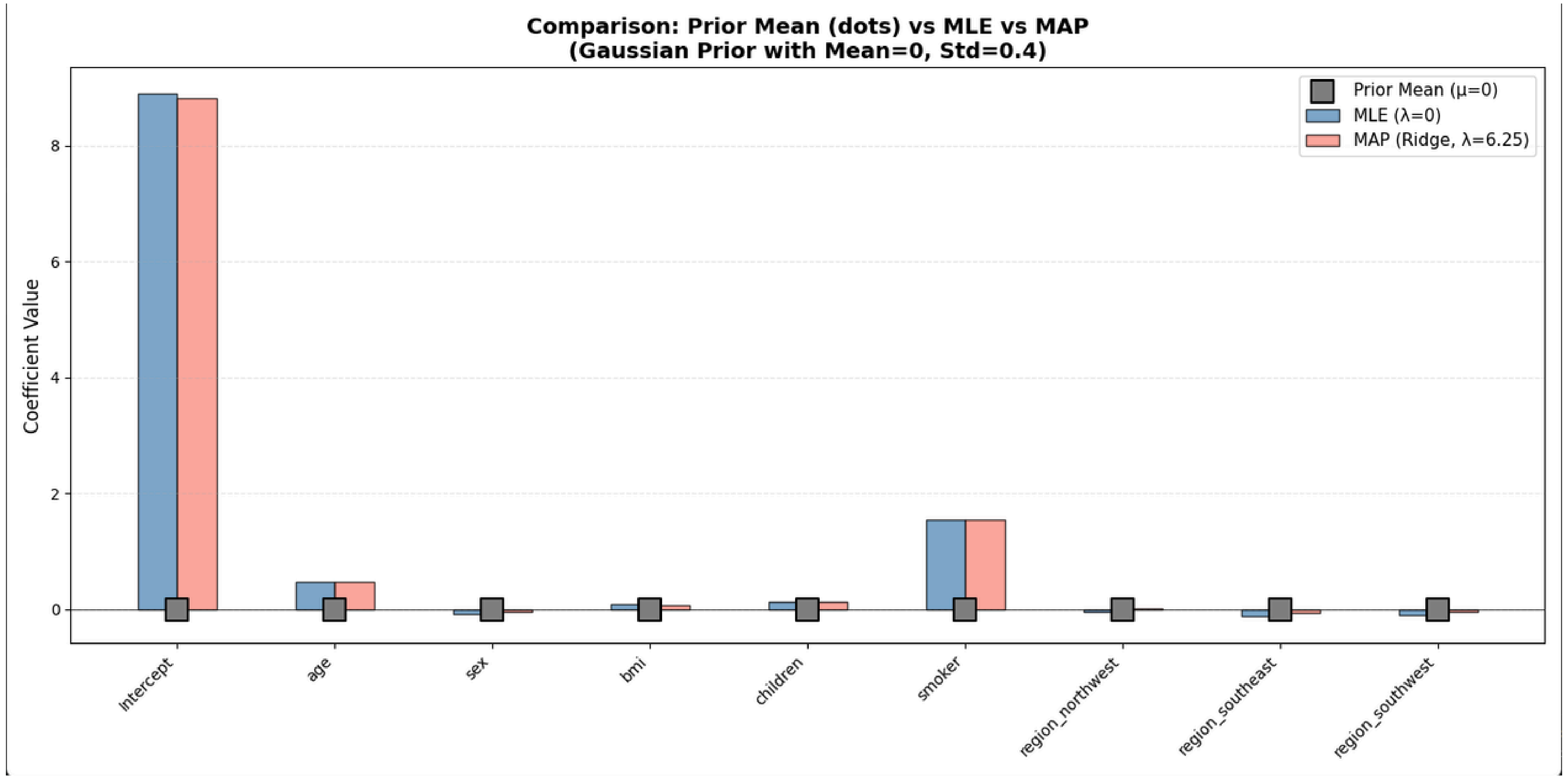


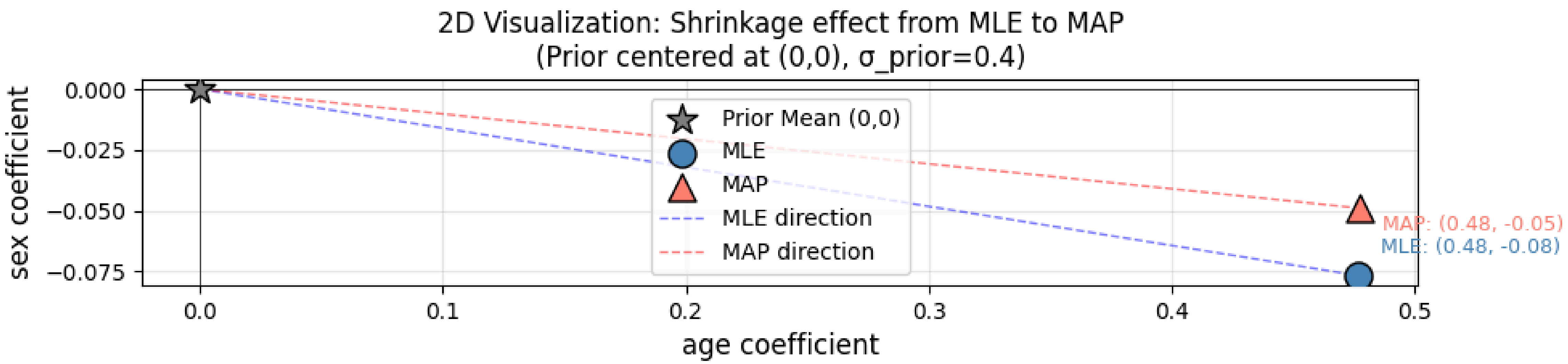

- Probabilistic Model: Gaussian Prior and MAP-based Log-Linear Regression for robust insurance cost modeling under stochastic noise.
- Optimization Strategy: Custom Gradient Descent for MAP objective minimization with Ridge-style regularization.
- Bayesian Integration: Combines observed data with prior knowledge ($\mu_{prior} = 0$) to produce noise-resistant calibrated predictions.

# Visual Analysis: MLE vs. MAP Results

- Compared to MLE, MAP is somewhat more 'hesitant' because it has a priori information that features do not affect the results.

- Sometimes this can help avoid overfitting.



# Conclusion

The project has successfully implemented the MAP estimation framework, proving it to be a powerful tool due to its ability to seamlessly integrate empirical data (Likelihood) with prior knowledge (Prior). Through three experimental problems, we draw the following core conclusions:

- Effective with small & noisy data: MAP (using Beta/Gaussian Priors) corrects the extreme estimates of MLE in small-sample coin flipping and filters out environmental noise in sensor calibration.
- Preventing Overfitting: In Linear Regression, MAP is equivalent to Ridge regression (L2 Regularization), which controls weight magnitudes and enhances the model's generalization capabilities.
- Stable Optimization: Utilizing the Negative Log-Posterior objective function combined with Gradient Descent provides a continuous, accurate, and efficient parameter optimization process.

**Takeaway Message:** In data science, integrating domain expertise (Prior) through MAP mathematics is the key to building robust and trustworthy AI models.